

B: Bus Assignment

Time Limit: 2 seconds, Memory limit: 2G

The Institution for Carrying People Carefully is responsible for managing the famous Line Bus in Line Town. The Line Bus goes through n stops conveniently numbered from 1 to n . At stop i , a_i people first get off the bus. Then, b_i people get on the bus. The bus starts out empty at stop 1 and then goes through the stops in numerically increasing order, eventually stopping at stop n where the bus empties.

When someone rides the Line Bus, they must be seated. A bus with capacity c has exactly c seats for passengers. Each rider of the Line Bus occupies exactly one seat. The driver of the Line Bus is not counted. The Institution for Carrying People Carefully wants to know what is the minimum capacity bus needed to run the Line Bus.

Input

The first line contains a single integer, n ($2 \leq n \leq 2 \cdot 10^5$).

Each of the next n lines contains two integers, a_i and b_i ($0 \leq a_i, b_i \leq 10^9$). It is guaranteed that at least one person boards the bus, at most 10^9 people board the bus over all stops, and that the bus will empty at stop n .

Output

Output a single integer, the minimum capacity bus needed to run the Line Bus.

Sample Input 1

```
4
0 3
1 2
2 1
3 0
```

Sample Output 1

```
4
```

D: Intuitive Elements

Time Limit: 1 second, Memory limit: 2G

Brandon is learning the periodic table! However, he doesn't like some of the elements because the symbol of the element contains letters which are not present in the name of the element. He finds this to be unintuitive, especially because in other contexts, he expects abbreviations to not introduce random letters.

Given a string and a proposed abbreviation, determine if Brandon would find it intuitive. Brandon finds an abbreviation intuitive if and only if every letter that appears in the abbreviation appears in the original string. Brandon does not look at the abbreviation carefully, so it is acceptable for a letter to appear more times in the abbreviation than in the original string, or for the letters to appear in a different order between the string and the abbreviation.

Input

The first line of input contains a single integer t ($1 \leq t \leq 10^3$). This is the number of test cases.

Each test case is represented on two lines.

The first line of each test case contains a single string a of length at least two and at most 50. This string only contains lowercase letters. The second line of the test case contains a single string b that is strictly shorter than a and also only contains lowercase letters.

Output

Output t lines, one for each test case.

For each test case, if all the letters in b appear in a , output YES. Otherwise, output NO.

Sample Input 1

```
4
magnesium
mg
silver
ag
aabb
bbb
aabb
ba
```

Sample Output 1

```
YES
NO
YES
YES
```

E: Mouse Pursuit

Time Limit: 3 seconds, Memory limit: 2G

Brandon is playing the newest idle game, Mouse Pursuit! The goal of this game is to pursue mice for cheese and glory.

In Mouse Pursuit, an *event* consists of pursuing a mouse. If the mouse is caught, the player might earn cheese and glory. However, if the mouse is not caught, the player might lose cheese and glory.

Given Brandon's recent events, Brandon wants to know how much cheese and glory he earned in the last k seconds.

Input

The first line of input contains a single integer, n ($1 \leq n \leq 10^5$).

The next n lines take one of two forms:

- **CAUGHT** s c g : A mouse was caught exactly s seconds ago. The player gained c pieces of cheese and g units of glory.
- **MISS** s c g : A mouse was missed exactly s seconds ago. The player lost c pieces of cheese and g units of glory.

For all events, $0 \leq c, g \leq 10^6$ and $1 \leq s \leq 10^9$. It is guaranteed that no two events happened at exactly the same time.

The last line contains a single integer k . It is guaranteed that no event happened exactly k seconds ago.

Output

Output two integers - the number of pieces of cheese Brandon gained in the last k seconds, and the number of units of glory Brandon gained in the last k seconds.

Sample Input 1

```
3
CAUGHT 1 6 5
MISS 4 1 2
CAUGHT 8 0 3
5
```

Sample Output 1

```
5 3
```

I: Exact Change

Time Limit: 1 second, Memory limit: 2G

While in Binaria, you find a store where you want to buy some presents for your friends. In Binaria, the currency is bits, and the coin denominations are the set of all integer powers of 2. You know that you want to spend at least a bits here, but no more than b bits.

When you make a purchase, you must pay with exact change. You have an unlimited number of bits that you can access from your bank account, but you can choose to withdraw them in whatever denominations you find most convenient. Carrying many coins is inconvenient though, so you wish to minimize the number of coins you carry with you.

Compute the minimum number of coins you need to bring with you such that you can pay any integer amount of bits between a and b , inclusive.

Input

The first line of input contains a single integer a , $1 \leq a < 2^{1000000}$. a will be written in base 2 with no leading zeros.

The second line of input contains a single integer b , $a \leq b < 2^{1000000}$. b will be written in base 2 with no leading zeros.

Output

Output a single integer k , the minimum number of coins you need to bring.

Sample Input 1

```
10101
101010
```

Sample Output 1

```
6
```

Sample Input 2

```
100
101
```

Sample Output 2

```
2
```